4(a)(i)	Any 2 from: • allows the reflected signal to be distinguished from the emitted signal • detection occurs in the time between emitted pulses • (reflection of ultrasound) detected by same probe / transducer / crystal • cannot emit and detect at same time (hence pulses)	**B2**
4(a)(ii)	piezo-electric crystal	**B1**
	ultrasound makes crystal vibrate / resonate	**B1**
	vibration produces (alternating) e.m.f. / p.d. across crystal	**B1**
4(b)(i)	$= (1.6 \times 10^6 - 4.3 \times 10^2)^2 / (1.6 \times 10^6 + 4.3 \times 10^2)^2$ = **0.999**	**B1**
4(b)(ii)	without the gel most of the ultrasound is reflected	**B1**
	Z values more similar / α reduces so less (ultrasound) is reflected / more (ultrasound) is transmitted	**B1**